

Writing a note

NSDF-Data student guide 3 — for every computer

Version 1 · 5 October 2026

Tip

What a note is. A note is a short, honest write-up of one question you investigated: what you asked, what you did, what you found, and how sure you are. Notes are how the lab keeps a record that someone else can follow and check. You do not need to be an expert to write one. You do need to be specific.

Before you start

- You have finished the setup guide and the first analysis for your computer.
- You know which question you are answering. If you do not, pick a small one from the project's *Next steps* sections (for example, a number you can reproduce from a note).
- You have a notebook (`.ipynb`) or a script that produces the numbers and figures you will show. Keep it. The note points to it.

What a note is for

A reader should be able to answer four questions after reading your note:

1. **What was the question?**
2. **What data and code did you use?**
3. **What did you find, and how sure are you?**
4. **What would you do next?**

If your note does not answer these, it is not finished, however long it is.

The parts of a note

Every note has the same parts, in this order. Each main part becomes a section with a heading and an outline link.

Section	What goes in it	Length
Title	The question, as a sentence. “Why does X happen in Y?” is better than “Analysis of X.”	one line

Section	What goes in it	Length
Header	Author, date, status, and one-sentence description (see below).	a few lines
Code repository	The notebook and library files you used, each pinned to the commit that produced the results.	a short list
Data and setup	Which series, dump and detector; which quantities; which windows and cuts. Define any word a newcomer would not know.	one or two paragraphs
The question	The question again, and why it matters.	a paragraph
Results	Your figures and tables, each with a caption that says what the reader is looking at.	as needed
Reading	What you think the results mean, and what they do <i>not</i> show.	a few bullets
Caveats	What could be wrong. Small sample? One channel only? A cut you chose by eye?	a short list
Next steps	Concrete things to do next.	a short list

Careful

Do not skip **Caveats**. Notes that state their limits are trusted. Notes without them are not, however good the plots look.

The header

Start your note file with a short block like this. These are the fields the site will read:

```

---
title: "Why does the 1000-sample window see more pulses?"
author: "Your Name"
date: "2026-10-05"
status: "In progress"
keywords: "baseline, pretrigger window, dump 1"
description: "One sentence: what you did and what you found."
---
```

Status is one of: *In progress* (you are still working on it), *Complete* (the question is answered and you are not going to change it), or *Outdated* (a later note or a fix has superseded it). New notes start

as *In progress*.

Writing the body

- **Say what you did in order**, with enough detail that someone could repeat it: the file, the detector, the window length, the cut.
- **Use the project's words**. A *segment* is a stretch of consecutive events with the same trigger type. A *run* is a whole data-taking period. Do not use *run* for a segment.
- **Define each quantity once**, in the *Data and setup* section, and use the same name every time after. If you call it `bstd` in one place and “baseline noise” in another, the reader will think they are different things.
- **Write short sentences**. Put the number first, then what it means: “The median is 1.53 counts, so a typical trace is flat noise.”
- **Say how sure you are**. “The two groups look the same” is weaker than “the medians are 1.51 and 1.58, from 141 and 38 traces.” Give the counts.
- **Separate what you measured from what you think**. A sentence in *Results* states a number. A sentence in *Reading* says what it might mean.

Tip

If you change your mind, say so. Notes that show a correction are more useful than notes that pretend nothing went wrong. Put the correction in a clearly labelled box, with the date.

Figures and tables

- **Every figure needs a caption** that says what is plotted, for which data, and what to look for. “Bstd against event number, dump 1. Grey: trigger type Unknown.” is a good caption.
- **Label the axes with units**. ADC counts, microseconds, or samples. Do not leave the units out.
- **Save figures as PNG** from your notebook, at a readable size, into the note's image folder.
- **Use a table** when you are comparing numbers. Use a figure when you are showing shape.

Pinning your code

A note is only useful if the reader can find the code that made its numbers. Link to your notebook and library files at a **specific commit**, not at the master branch, because master changes:

1. Commit your notebook to your branch (see the first analysis guide).
2. Find the commit's short code, for example `a1b2c3d`.
3. Link to https://github.com/villano-lab/NSDF-Data/blob/a1b2c3d/R76/analysis_notes/your-

Say in the note which commit you pinned.

Outline and links

Put a short outline near the top of your note, with a link to each main section. Readers use it to jump to what they need. Use one `##` heading for each main section, so that the outline can be built from your headings.

Checking your note before you submit

Go through this list before you ask for review:

- ☐ The title is a question.
- ☐ The header has author, date, status, keywords and description.
- ☐ Every number in the text can be found in your notebook output.
- ☐ Every figure has a caption with units and a sentence on what to look for.
- ☐ Every notebook link is pinned to a commit.
- ☐ Caveats are written, and you have said what you did *not* check.
- ☐ Next steps are concrete.
- ☐ You have read it aloud once, looking for a word a newcomer would not know.

How your note gets published

Careful

The publishing step is being set up. Until your lead confirms it is working, they will publish your note for you. Send them the note file, the image folder, and the commit you pinned.

When the workflow is ready, it will work like this:

1. You write your note as a Markdown file (a plain text file with simple formatting) in `notes/src/`. You can do this in the GitHub web editor, with no install.
2. You open a **pull request**: a request to add your change to the project. Your lead reviews it.
3. After the lead approves it, the site builds your note into a page and adds it to the index.

Whatever tool you use, you are responsible for the words and numbers in your note. If something you wrote turns out to be wrong, say so in the note and tell your lead.

Session Info

Guide version 1, 5 October 2026. Written for every computer. Please tell the project lead where a step was unclear.

Where to get help

Anthony Villano, project lead: anthony.villano@ucdenver.edu

When you write, include the note's title, the step or section you are stuck on, and the sentence or number that worries you. Nobody will be annoyed by a question about a note that is not finished.

Key links

- Notes site: <https://villano-lab.github.io/NSDF-Data/>
- Repository: <https://github.com/villano-lab/NSDF-Data>
- Example of a good note with a correction box: <https://villano-lab.github.io/NSDF-Data/notes/note-02a-bstd-vs-time.html>